

- 72 The table shows the DNA base triplet codes that are valid for this question and the amino acids that they code for.

<i>DNA base triplet</i>	<i>amino acid</i>
CGT	arginine
CGC	arginine
CGA	arginine
CGG	arginine
CTG	leucine
CCG	proline
CAG	glutamine
CAT	histidine
AGT	serine

Consider the part of the DNA sequence shown:

... CGC AGT ... →

Two mutations will occur in the DNA sequence. The initial mutation is an insertion of an additional base between the first and second bases in the sequence. The second mutation, which takes place some time after the first mutation, is a deletion that removes one base from the sequence.

Assuming that this DNA sequence is part of a longer sequence coding for a protein, and that no other mutations are occurring, what are the probabilities of the following?

	<i>probability that after the first mutation, the first triplet in this sequence does not code for arginine</i>	<i>probability that after the second mutation, both of the triplets in this sequence code for arginine</i>
A	$\frac{1}{4}$	$\frac{1}{28}$
B	$\frac{1}{4}$	$\frac{1}{24}$
C	$\frac{1}{4}$	$\frac{11}{28}$
D	$\frac{1}{4}$	$\frac{10}{24}$
E	$\frac{3}{4}$	$\frac{1}{28}$
F	$\frac{3}{4}$	$\frac{1}{24}$
G	$\frac{3}{4}$	$\frac{11}{28}$
H	$\frac{3}{4}$	$\frac{10}{24}$